6.5 Inequalities in One Triangle

Name _____ Period ____ Date _____

Geometry 6.5 · Student sheet

I can…

- list the angles of a triangle in order when I know the side lengths.
- list the sides of a triangle in order when I know the angle measures.
- decide whether three lengths can form a triangle and find the possible lengths of a third side.

Vocabulary · fill in the missing word

Opposite side (of an angle) — the side of the _____ that does not touch the angle

Triangle Longer Side Theorem — the angle _____ the longer side is larger

Triangle Larger Angle Theorem — the side _____ the larger angle is longer

Triangle Inequality Theorem — the sum of any two side _____ of a triangle is greater than the third side length

Part 1 · Discovery: measure and build

Measure. Use a ruler to draw a scalene triangle (all sides different). Measure each side and each angle. Circle the longest side and the largest angle.

Side	Length	Angle across from it	Measure

The largest angle is always _____

Build. Try to make a triangle with each set of strips.

Lengths (cm)	Triangle? (yes / no / lies flat)	Shorter two added
3, 4, 5		
3, 4, 9		
4, 5, 8		
3, 5, 8		
4, 4, 8		

Three lengths make a triangle when _____

Part 2 · Notes: fill in the blanks

Triangle Longer Side Theorem: If one side of a triangle is longer than another side, then the angle _____ the longer side is _____.

Triangle Larger Angle Theorem: If one angle is larger than another angle, then the side opposite the larger angle is _____.

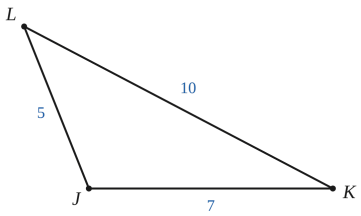
Triangle Inequality Theorem: The sum of the lengths of any two sides of a triangle is _____ than the length of the third side.

Third side: if two sides are a and b, then _____ < third side < _____

Part 3 · Examples (work them with the class)

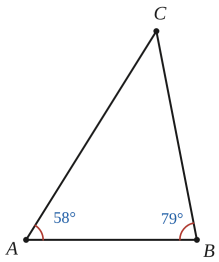
Example 1

List the angles of △JKL in order from smallest to largest.



Example 2

List the sides of △ABC in order from shortest to longest.



Example 3

Can the lengths form a triangle?

a. 4, 9, 12 b. 3, 6, 10 c. 5, 7, 12

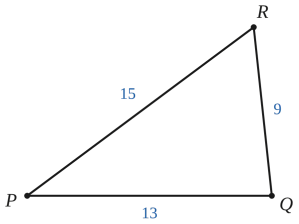
Example 4

A triangle has one side of length 12 and another side of length 8. Describe the possible lengths of the third side.

Part 4 · Partner practice

Partner 1

In $\triangle PQR$, $PQ = 13$, $QR = 9$, and $PR = 15$. List the angles from smallest to largest.



Partner 2

A triangle has sides of length 6 and 15. Describe the possible lengths of the third side.